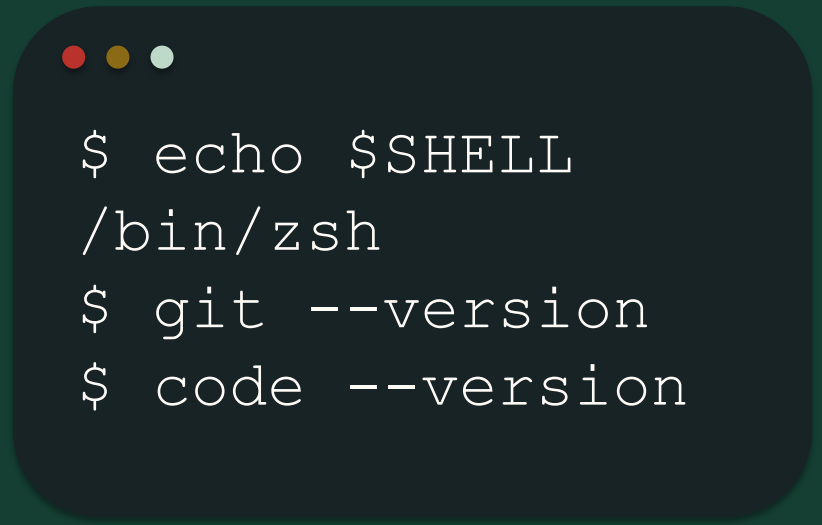
Developer Environment, Variables, Data Types

Module 0: Foundations



By the End of Today, You'll Be Able To…

1

Confirm your terminal, git, VS Code setup

2

Define variable, data type, and data structure

3

Sort 5 everyday items by data type

CASE STUDY

A Bad Deploy Cost $460 Million in 45 Minutes

What happened

New trading software + one old test flag → runaway trades

The cost

$460M lost in 45 minutes

Important point: Verify your setup with real validation steps from the start.

Source: U.S. Securities and Exchange Commission, Release No. 70694 (2013)

WHY THIS MATTERS

Setting Up Your Tools

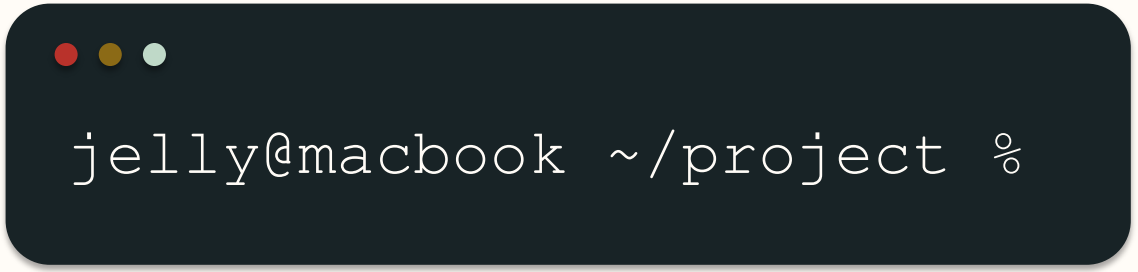


A setup you've actually verified beats one you're just assuming works — that habit starts now.

Reading Your Terminal Prompt



Bash-style prompt



Zsh-style prompt

Username

who you are

Machine

which computer

Folder

where you are

$ or %

ready for input

Git and VS Code Do Different Jobs

Git

A safety net for your code

Tracks changes so you can undo, compare, or combine work.

VS Code

Where you'll write your code

Your day-to-day code editor.

Both are free to download.

Code Along: Confirm Your Setup



Knowing what a tool is isn’t the same as knowing it actually works on your machine — let’s check.

Check Your Environment

bash

```
$ echo $SHELL
$ git --version
$ code --version
# confirm git knows who you are - commits need this
$ git config --global user.name
$ git config --global user.email
```

If a Command Isn’t Found

1

Read the actual error message first

2

Often means: not installed yet

3

Ask — don’t guess-and-retry repeatedly

Look Before You Type

bash

```
# open a real data file in a plain text view BEFORE
# running any inspection command on it
$ head -n 3 data/energy_sample.csv
```

Two Ways to Say Where a File Is

Relative

Starting from where you already are

`data/energy_sample.csv`

Absolute

The full path from the top

Starts with `/`

PATHS EXPLAINED

Code Along: Find Your Way Around

Relative and absolute are just words until you actually navigate with them.



Find Your Way Around

bash

```
$ pwd
$ ls
$ cd modules
$ ls
```

DISCUSS

Would You Rather?

Click

Navigate by clicking through folders

Type

Navigate with `pwd` / `ls` / `cd`

A Typo Looks Like This

bash

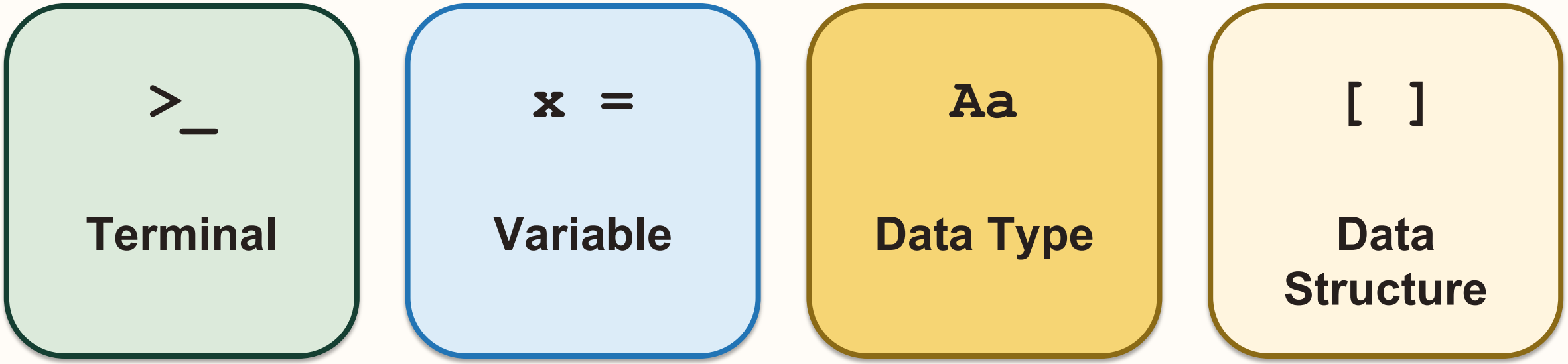
```
# a genuine typo - not a real command anywhere
$ gti status
```

The Vocabulary You'll Use Every Day

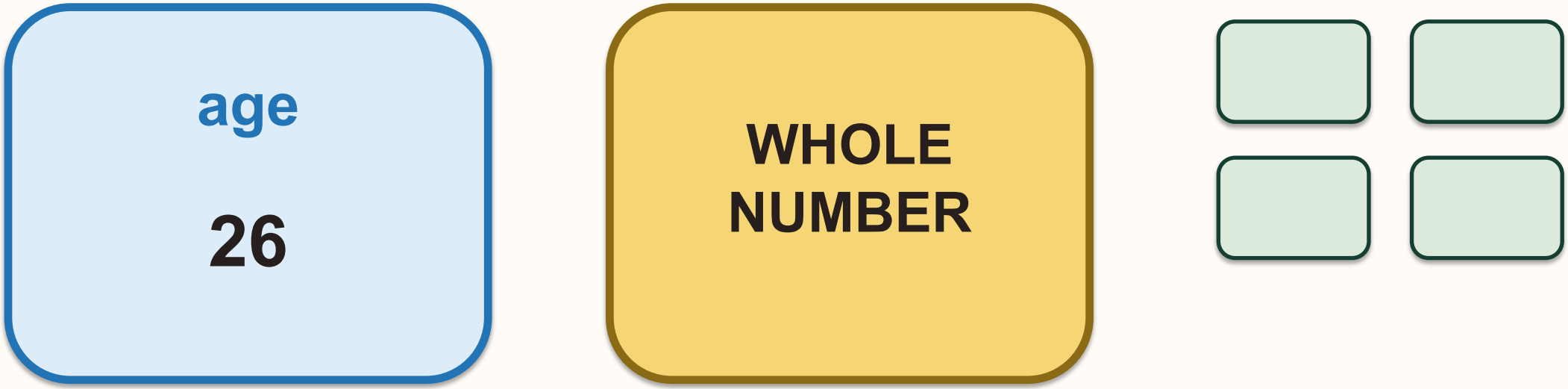


A working terminal doesn't tell you what you're actually looking at once you're in it — that's next.

Four Terms You'll Use Constantly



A Variable Is a Labeled Box



VARIABLE
name

DATA TYPE
what fits

DATA STRUCTURE
how values are organized

Five Data Types You'll See Constantly

Text	Whole number	Decimal	True / false	Date
name / address	count / age	price / temperature	in stock?	order / birthdate

Trap: ZIP codes and phone numbers look numeric, but they're text.

Put It Into Practice



Knowing a definition and recognizing it in the wild are two different skills.

CHECK FOR UNDERSTANDING

Shopping List: Type or Structure?

A

Data type

B

Data structure

ANSWER KEY

Shopping List: Type or Structure?

Correct answer: Data structure

A shopping list holds multiple items together, organized — not one single value.

QUICK CHALLENGE

Round-Robin: Name That Data Type

Name an everyday item. The room calls out its type.

A	B	C	D	E
Text	Whole number	Decimal	True / false	Date

ANSWER KEY

Round-Robin: Name That Data Type

Correct answer: Depends on the item named live

GPA → decimal • Wi-Fi password → text
Grocery list → data structure, not a single data type

Key Terms — Recap

Terminal

Where you type commands

Variable

A named value

Data Type

What kind of value

Data Structure

How values are organized

What You Have Now

✓

A working dev environment

✓

Real vocabulary for variable and data type

✓

The habit of looking before you type

Before Next Class

Keep today's setup

You'll reuse it.

Tomorrow

Git & GitHub — fully guided.

Git & Terminal Practice • Lab paired with the git lecture

KEEP LEARNING

Where to Go From Here

VS Code Docs

Getting Started
Official VS Code documentation

Software Carpentry

The Unix Shell
Hands-on terminal practice